

Day 21 Hybrid Notes

Measuring the outside and inside of solids | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can calculate surface area and volume of common solids.
I can choose the correct formula for prisms, cylinders, pyramids, cones, and spheres.
I can describe cross-sections created by slicing solids.

Part 1: Cloze Notes - Essential Ideas
1. Surface area measures the outside covering of a solid in _____.
2. Volume measures the space inside a solid in _____.
3. For a prism or cylinder, volume equals base area times _____.
4. For a pyramid or cone, volume is one third of base area times _____.
5. The volume of a sphere is four thirds pi r _____.
6. The surface area of a sphere is four pi r _____.
7. A cross-section is a two-dimensional shape formed by slicing a _____.

Part 2: Outline Notes - Big Picture A. Area formulas from Day 20 become base areas in _____. B. Surface area is useful for covering, wrapping, painting, or _____. C. Volume is useful for filling, capacity, storage, or _____. D. Cross-sections connect 3D objects back to familiar 2D _____. _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ _____
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 21 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. A rectangular prism is 10 by 6 by 4. Find volume and surface area.

Setup/work: _____

Answer: _____

2. A cylinder has radius 5 and height 12. Find volume.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student uses $V=Bh$ for a cone.

What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A box is 18 in by 12 in by 10 in. Find the cardboard needed to cover the outside, ignoring overlaps.

Setup: _____

Answer: _____

Sentence answer: _____